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Bio-Inspired Risk-Aware Adaptive Navigation for Quadruped Robots in Search and Rescue Scenarios

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Abstract

Search and Rescue (SAR) operations demand robotic systems capable of navigating hazardous and unstructured environments where human intervention is limited. This work introduces a bio-inspired navigation framework that enables quadruped robots to adapt their trajectories in real time based on perceived environmental risk. The approach integrates visual and depth perception with convolutional neural networks (CNN) to identify terrain features and estimate risk levels, generating dynamic maps that guide adaptive waypoint planning. A key aspect of this research is the extensive simulation campaign carried out in highly realistic post-disaster environments, including conditions with fire, smoke, debris, uneven surfaces, and degraded lighting. These scenarios provide a controlled yet challenging platform for validating adaptive behaviors before deployment in real missions. The proposed method is further evaluated in physical trials, confirming its ability to enhance safety, robustness, and efficiency compared to static navigation strategies. This study demonstrates the potential of combining bio-inspired principles with intelligent perception to enhance autonomous mobility in critical missions. The implemented algorithms and video of the testing phase are available in the Appendix.

Keywords: Bio-inspired navigation, adaptive locomotion, quadruped robots, search and rescue, risk-aware planning, sim-to-real transfer

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1. Introduction

Search and Rescue (SAR) operations present significant challenges due to the dynamic and unpredictable nature of disaster environments. Hazards such as fire, smoke, unstable surfaces, or debris, combined with limited accessibility, often constrain traditional human-led rescue efforts and increase operational risks [1, 2]. Quadruped robots have emerged as promising alternatives for these missions, offering superior mobility in rough terrains compared to wheeled or tracked platforms while reducing exposure for human responders. These robots can provide critical situational awareness through onboard sensing and support life-saving tasks, such as delivering supplies or locating victims in hazardous zones [3]. However, deploying such systems effectively requires more than robust locomotion; it demands adaptive navigation strategies capable of responding to rapidly changing conditions in real time.

Recent advances in reinforcement learning, model-based control, and terrain - aware planning have significantly improved quadruped autonomy. Nevertheless, real-world deployment remains challenging because disaster scenarios introduce high uncertainty and environmental variability. Factors such as degraded lighting, occlusions caused by smoke, and irregular ground surfaces can compromise perception and stability, leading to mission failure [4, 5, 6]. To address the challenges of unstructured environments, this work proposes a navigation framework rooted in the principles of *Visual Selective Attention* and *Anticipatory Locomotor Adaptation*. Central to this approach is the integration of Convolutional Neural Networks (CNN) that function as selective filters for environmental affordances. By fusing visual and depth data, the system constructs dynamic risk maps that categorize terrain based on traversability—distinguishing it from pure geometry. This semantic awareness drives a supraspinal feedforward control loop, where gait parameters such as velocity and swing height are modulated prior to physical interaction. Unlike traditional compliance controllers that rely solely on reactive spinal reflexes (feedback) after instability occurs, this architecture enables the robot to anticipate hazards and dynamically reconfigure its locomotion strategy, mirroring the predictive cognitive-motor integration observed in biological systems.

To ensure robustness before real-world deployment, an extensive simulation environment was developed to replicate post-disaster conditions with high fidelity in IsaacSim [4, 7]. The virtual scenarios include fire, smoke,

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debris, uneven terrain, and low-light conditions, providing a controlled yet challenging platform for validating adaptive behaviors. This simulation-first strategy not only accelerates development but also reduces the risks and costs associated with physical testing. The framework was subsequently validated in real-world experiments, demonstrating consistent improvements in navigation robustness and mission success compared to static planning approaches.

This work advances quadruped robot navigation for SAR missions through the following contributions:

- Bio-inspired adaptive navigation framework that dynamically adjusts waypoints based on environmental risk, enhancing robustness and tactical safety in unstructured terrains to support SAR missions without compromising operational effectiveness.
- Integration of convolutional neural networks for semantic risk detection, combining visual and depth data to generate dynamic risk maps that guide local trajectory adaptation in real time.
- High-fidelity simulation environment replicating post-disaster conditions—including fire, smoke, debris, uneven surfaces, and degraded lighting—to validate adaptive behaviors under realistic and challenging scenarios prior to physical deployment.
- Sim-to-real validation through experiments in controlled physical environments, demonstrating consistent improvements in navigation performance and mission success compared to static planning approaches.

The remainder of this work is organized as follows: Section 2 reviews related works applied to the SAR robotics context. Section 3 describes in detail the proposed framework, followed by results obtained and discussion in Section 4. Finally, Section 5 summarizes the conclusions and outlines future research directions.

2. Related Works

2.1. Biologically-Inspired Terrain Identification in Field Robotics

In Search and Rescue (SAR) missions, robust terrain perception is essential for safe locomotion in complex and unpredictable environments. Human perception relies predominantly on visual and depth cues to interpret

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geometry, texture, and distance, forming the basis for environmental understanding, learning experiences, and motion planning. Inspired by this, biologically-oriented approaches in robotics combine visual perception with proprioceptive and tactile feedback to estimate terrain affordances and associated risks more effectively, even under uncertain or changing conditions.

Antoun [8] pioneered non-visual sensing for disaster robotics through an echo-acoustic system inspired by cave-dwelling animals and visually impaired humans, enabling ultrasonic mapping and survivor heat detection. Building on internal sensing, Zhang et al. [9] proposed a proprioception-based terrain classification method using IMU and joint encoders, achieving motion-independent terrain labeling in terms of roughness, slippage, and slope—offering a low-cost and efficient alternative to external perception.

Cross-modal learning has further expanded terrain understanding. Chen et al. [10] introduced a self-supervised framework estimating terrain friction and stiffness from visual data, bridging simulation-trained policies with real-world performance. Similarly, Xu et al. [11] developed an unsupervised learning model inspired by animal cognition, enabling quadrupeds to infer terrain physical properties and resolve perceptual conflicts through associative tactile–visual layers.

Beyond passive sensing, Haddeler et al. [12] integrated active terrain probing with vision-based traversability analysis, formulating a collapsibility metric to assess non-rigid terrain safety. In parallel, Yu et al. [13] proposed a tapered whisker reservoir computing system for real-time terrain identification and adaptive control, demonstrating lightweight bio-inspired mechanical sensing.

To support benchmarking and fusion-based perception, Jeong et al. [14] released the DiTer dataset, combining RGB-D, thermal, LiDAR, and GPS data from quadruped experiments. Shu et al. [15] complemented this with a review highlighting the need for unified frameworks integrating semantic, geometric, and physical terrain features. Extending this, in previous works by the authors. [16, 17] presented a semantic probabilistic terrain analysis method using CNNs and Depth data to identify stable footholds and enhance adaptive locomotion in unstructured SAR environments.

2.2. Quadruped Locomotion and Planning Based on Risk Maps

Risk-aware locomotion planning is critical for quadruped robots operating in complex and evolving environments. While many methods rely on static

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terrain reconstructions, real-world conditions are dynamic, with debris, collapsible surfaces, and moving obstacles affecting traversability. Integrating perception-driven risk assessment into locomotion control allows robots to adapt to these changes and navigate safely.

Early approaches focused on safe foothold selection and robust motion over quasi-static terrain maps. Fankhauser et al. [18] proposed a real-time planner for ANYmal, computing collision-free swing-leg trajectories with continuous replanning, while Mastalli et al. [19] combined foothold, body motion, and step duration optimization using terrain cost-maps. Zhang et al. [20] integrated a Terrain-Aware Risk-Assessment Network into reinforcement learning, guiding policies to avoid risky actions, and Schneider et al. [21] used distributional Reinforcement Learning (RL) to generate explicit risk-sensitive locomotion behaviors.

Recent mapping strategies enhance terrain awareness by predicting traversability and hazard levels. Oh et al. [22] introduced TRIP, estimating steppability and risk online despite occlusions and data sparsity. Ashour et al. [23] built 3D semantic hazard maps, and Kantaros et al. [24] incorporated temporal logic planning under uncertain semantic maps. At the perception level, Zhang et al. [25] proposed the LSF-Planner, which uses RGB-D maps, surface features, and negative obstacle detection to construct a local risk-aware map, while Garcia Bermudez et al. [26] demonstrated terrain classification from proprioceptive and vibration data to inform locomotion.

High-level risk reasoning has also been explored. Shree et al. [27] fused visual data and natural language inputs to estimate danger levels for multi-robot SaR missions, generating trajectories that avoid high-risk areas while maintaining mission efficiency.

Although previous work has significantly advanced quadruped locomotion and terrain understanding, most approaches rely on static maps or pre-computed terrain models, which limits adaptability in dynamic and unpredictable disaster environments. While semantic and risk-aware planners improve foothold selection and local motion, few frameworks enable continuous, vision-driven adjustment of navigation strategies in response to evolving hazards. The approach presented in this paper overcomes these limitations by leveraging RGB-D perception to construct dynamic risk maps and guide adaptive planning in real time, allowing quadruped robots to adjust their trajectories without modifying the underlying locomotion controller.

3. Methodology

3.1. System Overview

The proposed framework is structured into four main stages, as illustrated in Figure 1. The first stage focuses on the simulation environment, where high-fidelity post-disaster scenarios are reconstructed to replicate real-world conditions such as fire, smoke, debris, and degraded lighting. To ensure physical realism, the virtual environments incorporate diverse surface textures and assign specific friction coefficients to ground elements, enabling accurate modeling of locomotion dynamics and slip conditions. This approach provides a controlled yet challenging platform for validating navigation strategies under representative SAR conditions before physical deployment.

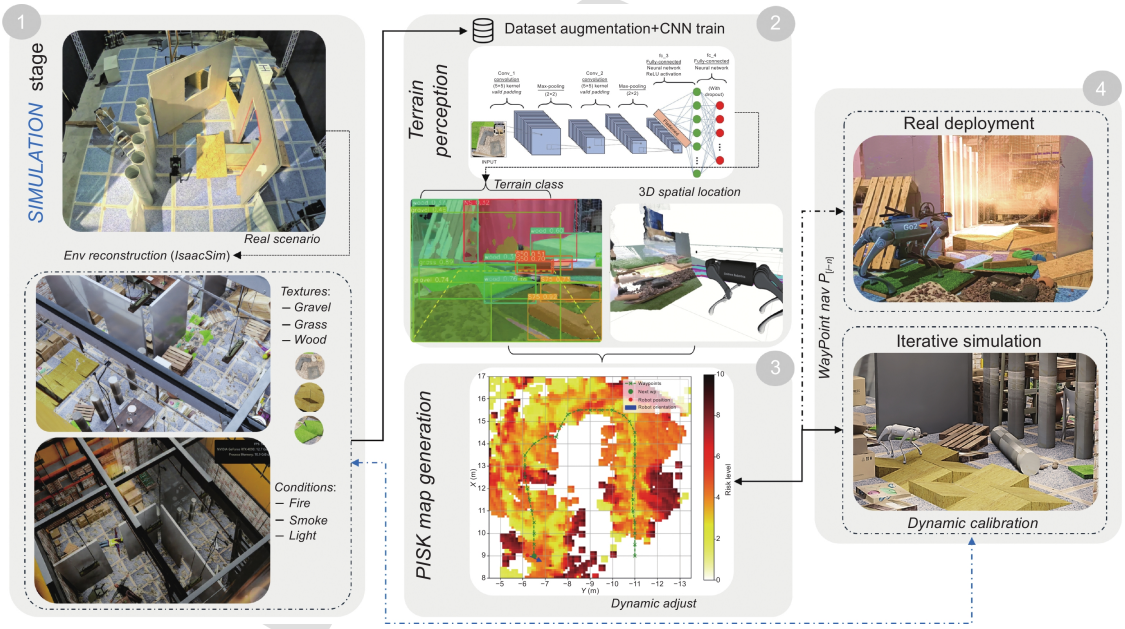


Figure 1: Overview of the proposed method structured in four main stages.

The second stage addresses terrain perception and risk estimation. A convolutional neural network processes RGB and depth data to classify terrain types and obstacles, assigning risk levels based on semantic context. These detections are processed with elevation information and fused into a dynamic risk map (third stage), which serves as the basis for adaptive local waypoint

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planning towards a final destination area (global waypoint). The final stage involves deployment, first through simulation and then in the real environment. The mapping strategies perfected during simulation are ultimately transferred to the robot for real-world SAR application..

3.2. Simulation Environment: Requirements and Implementation

The environment was implemented in IsaacSim 4.2, selected for its advanced physics engine and photorealistic rendering capabilities. The design adhered to the following requirements:

1. **Realistic Terrain Modeling:** The simulation environment replicates structural collapse and irregular surfaces using photorealistic textures and calibrated static and dynamic friction coefficients (μ_s, μ_d). These values were selected from empirical ranges corresponding to disaster-related materials and validated against NIST standards:

$$\mu_s \in [0.30, 0.65], \quad \mu_d \in [0.25, 0.55].$$

2. **Hazard Representation:** Dynamic hazards such as fire, smoke, and degraded lighting are parameterized by intensity L and particle density H:

$$L \in [0.2, 1.0] \text{ (normalized)}, \quad H \in [0.1, 0.8] \text{ (relative concentration)}.$$

3. **Sensor Deployment Effects:** RGB and Depth sensors mounted on the quadruped introduce motion-induced noise σ_n and vibration artifacts, modeled as Gaussian perturbations:

$$\sigma_n \approx 0.02 \text{ m (depth)}, \quad \text{frame jitter} \approx 5^\circ.$$

4. **Structural Similarity (SSIM):** Quantifies perceptual similarity between simulated and real images ($0 \leq \text{SSIM} \leq 100$).

Figures 2(a)-(c) illustrate the progression from physical test arenas to advanced virtual scenarios incorporating fire, smoke, and low-light conditions. The baseline physical arena (Figure 2(a)) includes obstacles, and confined spaces to emulate structural collapse, serving as a reference for validating virtual fidelity. Its simulated counterpart (Figure 2(b)) reproduces debris and irregular surfaces with photorealistic textures and calibrated friction coefficients (μ_s, μ_d), ensuring accurate modeling of terrain variability and slippage.

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Finally, Figure 2(c) presents an advanced configuration with fire, smoke, and degraded lighting, creating a highly challenging environment for perception systems and enabling the assessment of CNN-based semantic detection under visual degradation and dynamic hazards.

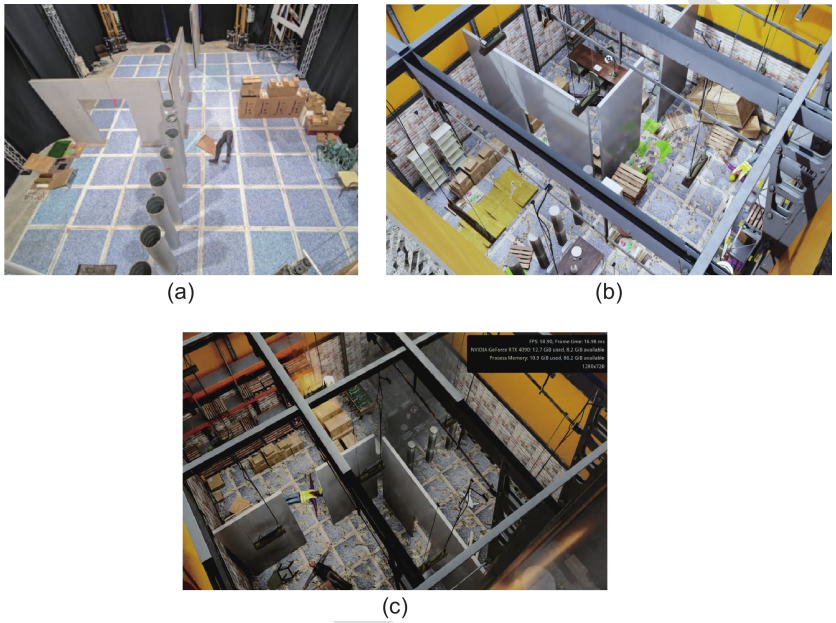


Figure 2: Environments for validation, (a) the physical test scenario, (b) a simulated environment with debris and structural clutter, and (c) the simulated environment under degraded visibility conditions, ensuring coverage of both nominal and adverse operational contexts.

3.2.1. Environmental Calibration and NIST Compliance

To ensure that the simulated hazards represent realistic post-disaster conditions, the environmental parameters—Light Intensity (L) and Smoke Particle Density (H)—were calibrated following the *NIST Standard Test Methods for Response Robots* (ASTM E2828) [28]. The simulation utilizes Isaac Sim's ray-tracing engine to model physical light transport and volumetric scattering.

The degradation of visual perception is parameterized as follows:

- **Lighting (L):** Defined as the ambient illuminance in Lux (lx). To replicate "degraded lighting" scenarios defined by NIST, L is varied

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from standard operational lighting (> 300 lx) down to blackout conditions (< 1 lx).

- **Smoke Density (H):** Defined by the volumetric extinction coefficient σ_e (m^{-1}), which correlates to the particle density. The effective visibility distance V_{vis} is derived using Koschmieder's Law [29]:

$$V_{vis} = \frac{3.912}{\sigma_e(H)} \tag{1}$$

where H modulates the opacity of the particle system.

Table 1 summarizes the parameter ranges used to train and validate the system, ensuring that they cover the operational domains specified for urban search and rescue.

Table 1: Simulation Parameters and NIST Correlation.

Condition	**Intensity L (Lux)**	**Density H (σ_e)**	**NIST/Real-world Equivalent**
Clear	> 300	0.0	Standard Office / Daylight
Low Light	$10 - 50$	0.0	Twilight / Failed Power Grid
Haze/Dust	$100 - 300$	$0.1 - 0.5$	Post-collapse concrete dust ($V_{vis} \approx 8 - 40$ m)
Dense Smoke	< 50	$1.0 - 5.0$	Active Fire Proximity ($V_{vis} < 4$ m)
Extreme	< 5	> 5.0	Zero Visibility (Thermal Cam Required)

By varying L and H within these physically grounded ranges, we ensure that the CNN-based risk estimator is robust against the optical distortions encountered in physical deployment.

3.3. Risk-Aware Environment Understanding

Robust navigation in post-disaster environments requires an integrated understanding of terrain semantics and spatial constraints. The implemented method involves RGB-based semantic detection with depth-based spatial reasoning to emulate human decision-making, where stability and traversability are prioritized over visual appearance.

3.3.1. Semantic Context

Semantic perception relies exclusively on RGB images ($\mathcal{I}_t \in \mathbb{R}^{H \times W \times 3}$) processed by the CNN (YOLOv8n-seg) architecture. Selected for its balance between inference speed and accuracy on edge hardware, the model utilizes

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a CSPDarknet53 backbone with C2f modules and processes inputs at a resolution of 640×640 pixels. The network was trained on a custom dataset of 10,835 images (70% train, 20% validation, 10% test) comprising nine classes relevant to terrain identification: `concrete`, `grass`, `gravel`, `wood`, and stable locomotion `NS` (non-surmountable), `S50`, `S75`, `S100` (ground obstacles with a 50-75-100% probability of being overcome, defined based on a human criterion in the labeling phase), and also `victim`.

To ensure robustness against the visual degradation typical of SAR scenarios, training employed the SGD optimizer (momentum=0.937, weight decay=0.0005) with an initial learning rate of 0.01 and a batch size of 16, running for 259 epochs (converging at epoch 159). Furthermore, a specific augmentation pipeline was applied, including Gaussian blur (max 1.5px), random noise injection (0.5%), and random rotations ($\pm 5°$), alongside standard mosaic augmentation. These classes were selected to reflect variations in texture, friction, and slope critical for stability (available in the Appendix C).

Figure 3(a) shows the Precision-Recall (training performance) curves for all classes at IoU=0.5. The overall mean Average Precision (mAP) reached 0.740, indicating robust detection across most classes. Notable observations include:

- **High performance:** `grass` achieved the highest AP (0.962), followed by `gravel` (0.892) and `S100` (0.891), reflecting strong detection for stable and fully surmountable surfaces.
- **Moderate performance:** `concrete` (0.834) and `victim` (0.746) showed reliable detection but room for improvement.

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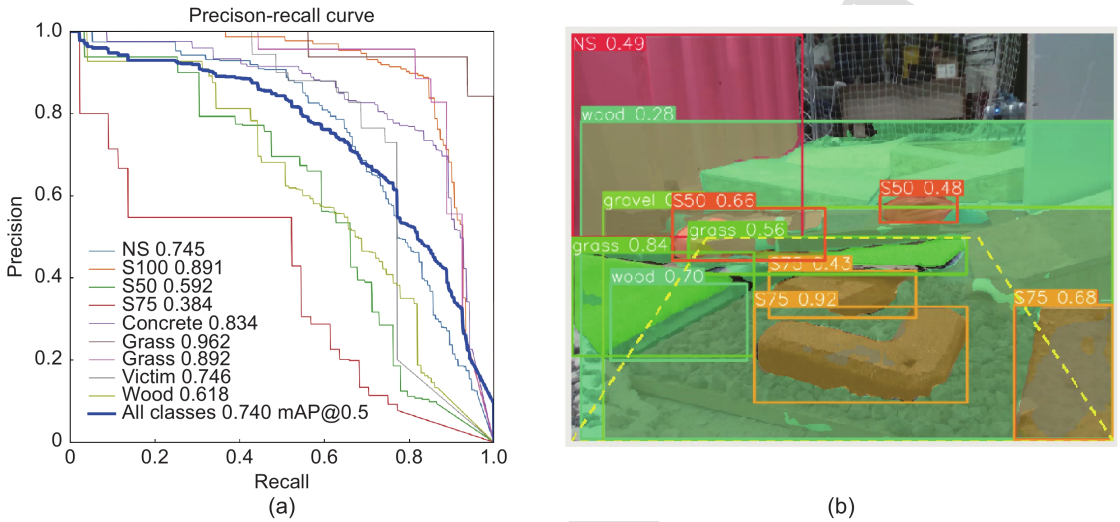


Figure 3: (a) Precision-Recall curves for terrain and structural classes. Example of semantic inference in the real environment.

Figure 3(b) shows semantic inference in the real environment. The RGB frame is processed by the CNN to detect terrain classes and structural elements relevant to locomotion. Bounding boxes indicate predicted classes and confidence scores: `NS` (non-surmountable), `wood`, `gravel`, `grass`, and slope categories (`S50`, `S75`). These detections feed the risk mapping process by assigning predefined risk scores $r(c_k)$ to each class, which are integrated into the grid $\mathcal{G}$ for adaptive trajectory planning. The stability criteria underlying this classification follow principles previously studied in [16], where terrain semantics and locomotion safety were analyzed for legged robots in unstructured environments.

3.3.2. Spatial Context

Depth maps ($\mathcal{D}_t \in \mathbb{R}^{H\times W}$) complement semantic detection by providing ground metric distances, enabling projection of CNN detected objects into the world frame (ref. Figure 4(a)). For each detection $z_k = (c_k, \mathbf{p}_k, s_k)$, where c_k is the class, $\mathbf{p}_k = (u_k, v_k)$ the pixel coordinates, and s_k the confidence score, the depth at (u_k, v_k) is d_k. The global position is computed following 2:

$$\mathbf{x}_k^W = \mathbf{T}_{WR}(t)\,\mathbf{T}_{RC}\begin{bmatrix} d_k K^{-1}[u_k, v_k, 1]^\top \\ 1 \end{bmatrix}, \tag{2}$$

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where K is the intrinsic matrix, and $\mathbf{T}_{WR}, \mathbf{T}_{RC}$ are the world-to-robot and robot-to-camera transforms.

3.4. Risk Map Generation

The semantic and spatial information obtained from the previous stage is integrated into a dynamic risk representation that guides adaptive navigation (The implementation algorithms are available in Appendix A). The environment is discretized into a grid covering the operational area with metric coordinates:

$$\mathcal{G} \in \mathbb{R}^{M \times N}, \quad \text{resolution } \rho \text{ (m/cell)},$$

For each detection $z_k = (c_k, \mathbf{p}_k, s_k)$, the semantic class c_k is mapped to a predefined risk score $r(c_k)$ according to the Table 2 derived from iterative simulations stability criteria:

Table 2: Risk scores for terrain types.

Label	**Risk Score**	**Label**	**Risk Score**
concrete	2	S50	6
wood	9	S75	8
grass	5	S100	4
gravel	8	NS (Non-surmountable)	10

Classes such as `NS` (non-surmountable) and steep slopes (`S75`) receive higher risk values, while stable surfaces like `ground` or `concrete` are assigned lower scores. This mapping emulates human-like judgment of traversability in unstructured terrains.

The risk values were selected to prioritize stability and safety during navigation, following principles observed in state-of-the-art works on legged locomotion in unstructured environments [16, 30]. Risk values placed over the map (Figure 4(b)) are distributed over a local neighborhood using a Gaussian kernel:

$$\mathbf{K} = \begin{bmatrix} 0.8 & 0.9 & 0.8 \\ 0.9 & 1.0 & 0.9 \\ 0.8 & 0.9 & 0.8 \end{bmatrix},$$

which mitigates abrupt transitions and accounts for uncertainty in detection.

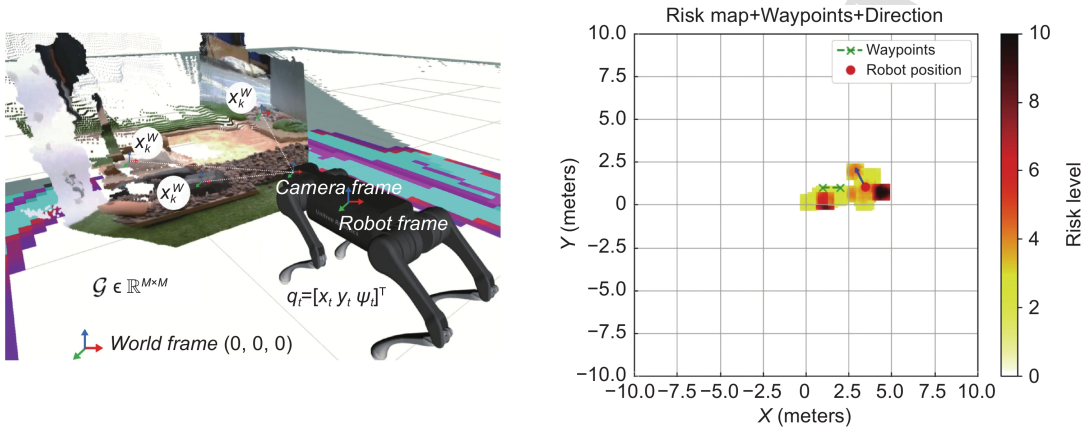


Figure 4: (a) 3D visualization of the perception system of the robot in the operating environment. (b) Initial risk map generated during navigation. Risk values are smoothed using a Gaussian kernel to mitigate abrupt transitions. A thermal colormap is used, ranging from white and yellow (low risk), through orange and red (moderate to high risk), up to black (extreme or non-traversable zones).

Each cell $\mathcal{G}$ stores two cumulative statistics: the risk sum $\mathbf{S}(i,j)$ and the observation count $\mathbf{N}(i,j)$. Depth information $\mathcal{D}_t$ complements semantic detection by providing metric distances and height variations, enabling accurate projection of detected objects into the global map and computation of relative elevation for risk aggregation.

- $\mathbf{S}(i,j)$ stores the **risk sum**, i.e., the cumulative magnitude of risk observed in cell (i,j).
- $\mathbf{N}(i,j)$ stores the **observation count**, i.e., the number of times the cell has been updated.

The mean risk per cell, incorporating elevation, is computed as equation 3:

$$\bar{\mathbf{R}}(i,j) = \begin{cases} \dfrac{\mathbf{S}(i,j) + \lambda \cdot |\Delta h(i,j)|}{\mathbf{N}(i,j)}, & \mathbf{N}(i,j) > 0, \\ \text{NaN}, & \text{otherwise.} \end{cases} \tag{3}$$

Here, $\Delta h(i,j)$ represents the elevation difference relative to the terrain baseline, and λ is a weighting factor.

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3.5. Adaptive Trajectory Adjustment and Locomotion

Robust navigation in hazardous environments requires coupling risk-aware trajectory planning with adaptive quadruped locomotion. The proposed approach integrates dynamic waypoint adjustment based on the risk map with a locomotion controller that modulates gait parameters according to terrain conditions.

3.5.1. Risk-Aware Waypoint Update

The nominal path $\mathcal{W} = \{\mathbf{w}_1, \ldots, \mathbf{w}_n\}$ is refined online using the risk map $\hat{\mathbf{R}}_t$. Each waypoint—corresponding to an area of interest—is evaluated. If its local risk exceeds a threshold, the system identifies a safer neighboring cell. However, to prevent unreachable targets, a *feasibility check* is performed before applying the shift (Algorithm 1).

Algorithm 1 Feasibility-Aware Waypoint Adjustment

Require: Waypoint $\mathbf{w}_i$, risk map $\hat{\mathbf{R}}_t$, threshold τ
$r_{\text{local}} \leftarrow \text{MeanRisk}(\hat{\mathbf{R}}_t, \mathbf{w}_i, \delta)$
if $r_{\text{local}} > \tau$ **then**
  $\mathbf{p}_{cand} \leftarrow \text{FindLowestRiskNeighbor}(\mathbf{w}_i)$ ▷ Feasibility Check
  if $\text{CheckFeasibility}(\mathbf{p}_{cand})$ **then**
    $\mathbf{w}_i \leftarrow \text{MoveToward}(\mathbf{p}_{cand}, \text{step_size})$
  end if
end if
$\mathcal{W}^* \leftarrow \text{UpdateSet}(\mathbf{w}_i)$
$\mathcal{W}_{smooth} \leftarrow \text{BSplineInterpolation}(\mathcal{W}^*)$ ▷ Ensures continuity
return $\mathcal{W}_{smooth}$

The validation function CheckFeasibility ensures the candidate point $\mathbf{p}_{cand}$ is within the kinematic workspace ($d < d_{max}$) and the local slope is traversable ($\phi < 30°$).

Spatial Risk Evaluation and Trajectory Smoothing. While equation 3 defines the temporal risk accumulation per individual cell, the local planner requires a spatial assessment to ensure safety for the robot's entire footprint. The MeanRisk function used in Algorithm 1 computes the spatial average within a safety radius δ around a candidate waypoint $\mathbf{p}$:

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$$\text{MeanRisk}(\mathbf{p}, \delta) = \frac{1}{|\mathcal{N}_\delta(\mathbf{p})|} \sum_{(i,j)\in\mathcal{N}_\delta(\mathbf{p})} \bar{\mathbf{R}}(i,j) \tag{4}$$

where $\mathcal{N}_\delta(\mathbf{p})$ is the set of grid indices within distance δ of $\mathbf{p}$.

However, adjusting waypoints individually based on local risk can lead to discontinuous paths if the correlation between consecutive points is neglected. To address this, the set of risk-adjusted waypoints $\mathcal{W}^*$ serves as the control polygon for a cubic *B-Spline interpolation*. The final continuous trajectory $\mathbf{P}(u)$ is generated as:

$$\mathbf{P}(u) = \sum_{i=0}^{n} \mathbf{w}_i^* N_{i,k}(u) \tag{5}$$

where $N_{i,k}(u)$ are the basis functions of degree $k = 3$. This formulation enforces geometric correlation between adjacent waypoints and guarantees C^2 continuity, ensuring that the velocity and acceleration commands sent to the locomotion controller remain smooth even after aggressive local risk avoidance.

3.5.2. Locomotion Control Hierarchy

The locomotion stack consumes the updated waypoints and local risk context through a three-tiered architecture, ensuring that high-level decisions are dynamically translated into stable whole-body motions:

- **High-level Navigation (10 Hz):** This layer generates the desired velocity commands $\mathbf{u} = [v_x, v_y, \omega_z]^T$ using a "turn-then-go" strategy to align the robot with the safe path. The heading error is corrected via proportional control $\omega_z = k_\psi(\psi_d - \psi)$, while the linear velocity vector $[v_x, v_y]$ is proportional to the distance to the target Δp. Crucially, the maximum speed is dynamically scaled by the local risk factor to enhance stability in hazardous zones.

- **Mid-level Gait Planner (100 Hz):** This module selects the appropriate gait pattern (e.g., trot or crawl) and modulates swing trajectory parameters based on terrain semantics. To avoid tripping hazards, the swing height h_{swing} is adjusted according to the risk score r^* of the intended foothold, as defined in equation (6):

$$h_{swing} = h_0(1 + \gamma r^*) \tag{6}$$

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where h_0 is the nominal clearance and γ is a gain factor. Foothold candidates are further scored and filtered based on local slope and kinematic reachability.

- **Low-level NMPC (100 Hz) & WBC (500 Hz):** The motion tracking relies on a Nonlinear Model Predictive Control (NMPC) running at 100 Hz. The state vector is defined as $\mathbf{x} = [\mathbf{h}_{com}^T, \mathbf{q}_b^T, \mathbf{q}_j^T]^T$, comprising the normalized centroidal momentum $\mathbf{h}_{com} \in \mathbb{R}^6$, base coordinates $\mathbf{q}_b$, and joint positions $\mathbf{q}_j$. The control input $\mathbf{u} = [\mathbf{f}_c^T, \mathbf{v}_j^T]^T$ consists of contact forces and joint velocities.

  The controller solves the optimization problem formulated in equation (7) over a finite horizon:

$$\min_{\mathbf{u}(\cdot)} \int_{t_0}^{t_f} \left(||\mathbf{x} - \mathbf{x}^{ref}||_{\mathbf{Q}}^2 + ||\mathbf{u} - \mathbf{u}^{ref}||_{\mathbf{R}}^2 \right) dt \tag{7}$$

  subject to system dynamics $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$ and constraints (friction cones, kinematic limits). The optimized trajectories are tracked by a Whole-Body Controller (WBC) running at **500 Hz**, which prioritizes tasks—such as friction constraints and base tracking—to compute final joint torques.

3.6. Experimental Setup and Evaluation Metrics

Experiments were conducted in an indoor environment following the NIST Orange Zone standard for moderate complexity (ramps, clutter, industrial-like conditions) [31, 32]. Tests included high-fidelity simulation in *Isaac Sim* and real-world validation with a Unitree quadruped robot. Table 3 summarizes the conditions and key parameters.

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Table 3: Experimental Setup Summary

Aspect	Details
Environment	Indoor scenario (Isaac Sim + CAR-Robotics-Arena), compliant with NIST Orange Zone (moderate obstacles, ramps, clutter).
Obstacle Density	Three levels: Low, Medium, High.
Coverage Area	Straight path (nominal) and mobility strategies (coverage-oriented for map generation).
Locomotion Modes	Uninformed (no risk data), Semi-informed (post-failure adjustments), Informed (real-time risk-aware adaptation).
Tests Conducted	Map generation under normal and challenging conditions; evaluation of locomotion strategies.
Real Validation	Unitree Quadruped robot in CAR-Robotics-Arena indoor arena replicating simulated conditions.
Evaluation Metrics	Success rate, mission time, path deviation, risk map coverage.

3.6.1. Definition of Evaluation Metrics

To provide a rigorous quantitative assessment, the evaluation metrics listed in Table 3 are mathematically defined as follows:

- **Trajectory Completion (TC):** This metric quantifies the locomotion progress relative to the planned route. Specifically, it refers to the **percentage of the planned path length** successfully traversed by the robot, rather than a simple binary success/failure status. It is calculated as:

$$TC(\%) = \frac{L_{traveled}}{L_{ref}} \times 100 \tag{8}$$

  where $L_{traveled}$ is the arc length covered and L_{ref} is the total reference path length.

- **Success Rate (SR):** Derived from the previous metric, a trial is classified as a "Success" if the robot reaches the goal zone without critical failures (collision, entrapment). Mathematically, this corresponds to a Trajectory Completion of $TC \geq 95\%$.

- **Risk Map Coverage (C_{map}):** This metric represents the **proportion of the environment effectively mapped by the risk map**, defined as the ratio of grid cells populated with valid semantic risk scores (N_{risk})

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relative to the total traversable area (N_{total}):

$$C_{map}(\%) = \frac{N_{risk}}{N_{total}} \times 100 \tag{9}$$

It distinguishes active semantic perception from simple geometric exploration.

- **Mission Time (T) and Path Deviation (e_{traj}):** T denotes the total duration to reach the goal or trigger a failure. e_{traj} measures the tracking fidelity as the Root Mean Square Error (RMSE) of the robot's position relative to the optimal path.

3.6.2. Computational Cost Analysis

To ensure the viability of the proposed framework in dynamic scenarios, the computational cost of the critical navigation modules was evaluated on the NVIDIA RTX 4090 workstation. As evidenced by the performance graph in Figure 2(c), the integrated simulation environment maintains an average frame rate of 58.9 FPS (approximately 17 ms per frame), validating the efficiency of the implementation.

The total latency was analyzed by decomposing the execution time of the primary threads:

- **CNN Inference:** The YOLOv8n-seg model (input 640×640) achieved an average inference rate of 29 FPS (approximately 34.5 ms latency). Running on a parallel thread, this creates a perception update rate that comfortably exceeds the 10 Hz ($\sim$100 ms) requirement for high-level decision-making.

- **Map Updates:** The aggregation of semantic measurements into the dynamic risk grid ($\mathcal{G}$) is optimized to process depth point clouds locally, resulting in an average execution time of ≈ 5 ms per cycle.

- **Waypoint Replanning:** The local adjustment algorithm operates on a constrained search window (3×3 neighborhood). The computational cost is negligible, requiring < 0.1 ms to compute the optimal safe neighbor.

Consequently, the combined computational load ensures that perception and planning tasks do not bottleneck the high-frequency stability controllers.

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4. Results

4.1. Quantitative Assessment of Sim-to-Real Environment Consistency

The structural similarity of the simulated environment was assessed through feature-based correspondence analysis against its real-world counterpart. High - resolution images were preprocessed using morphological operations to normalize structural details, followed by keypoint detection and description with SIFT and SURF. The resulting correspondences concentrated on critical geometric features such as stair edges, risers, and junctions, confirming that the simulation preserves invariant local patterns essential for perception-driven navigation. Quantitatively, the concordance of detected features reached 79% for Case - (a) and 87% for Case - (b), indicating a high degree of structural and textural consistency between the simulated and real environments.

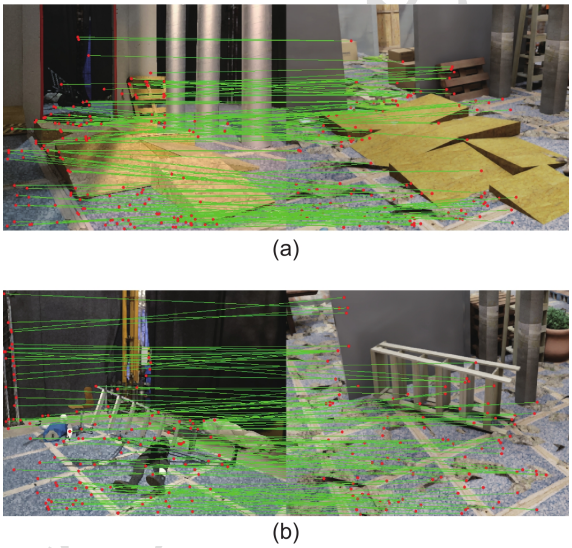


Figure 5: Feature correspondences between real (left column) and simulated (right column) environments. Green lines indicate matched keypoints; red dots mark detected features.

Impact on Transferability and Mission Reliability. Validating structural fidelity is a critical prerequisite for *sim-to-real transferability*, effectively mitigating the *domain gap* inherent in deep learning models. The observed feature correspondence (79%–87%) confirms that invariant geometric descriptors persist across domains despite variations in lighting and texture. This

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ensures that the risk assessment logic trained virtually generalizes robustly to physical deployment without requiring extensive fine-tuning.

For SAR mission reliability, this consistency is paramount. In unstructured disaster scenarios, perceptual aliasing caused by domain discrepancies—such as misclassifying real smoke or unstable debris due to texture artifacts—can lead to catastrophic locomotion failure. Validated simulation fidelity ensures that anticipatory avoidance behaviors will be triggered reliably when the robot encounters analogous hazards in the real world.

4.2. Risk Map Generation

4.2.1. Obstacle Density

Figure 6 illustrates risk maps generated in environments with low, medium, and high obstacle density. In all cases, the robot incrementally updated the map during navigation, applying Gaussian kernel smoothing to avoid abrupt risk transitions. For low density (a), risk distribution remains sparse, with clear safe corridors. Medium density (b) introduces localized high-risk clusters near obstacle groups, requiring minor waypoint adjustments. High density (c) produces widespread elevated risk, significantly constraining safe paths and increasing the complexity of adaptive planning. Despite these variations, the mapping process maintained stability, confirming the robustness of the risk aggregation strategy under progressively challenging conditions.

4.2.2. Coverage of the environment

The effect of mobility strategy on risk map coverage was analyzed under high obstacle density conditions. Two strategies were compared: normal straight-line navigation (Figure 6(d)) and zigzag coverage-oriented movement (Figure 7). Each experiment was repeated ten times, and the values reported in Table 4 correspond to the mean results.

Table 4: Comparison of Normal vs Zigzag Strategy under High Obstacle Density.

Strategy	**Mean Time (s)**	**Mean Coverage (%)**
Normal	102	69.64
Zigzag	147	91.65

The zigzag strategy increased risk map coverage by approximately 31.6% relative to normal navigation (91.65% vs. 69.64%), demonstrating its effectiveness for comprehensive mapping. However, this improvement came at

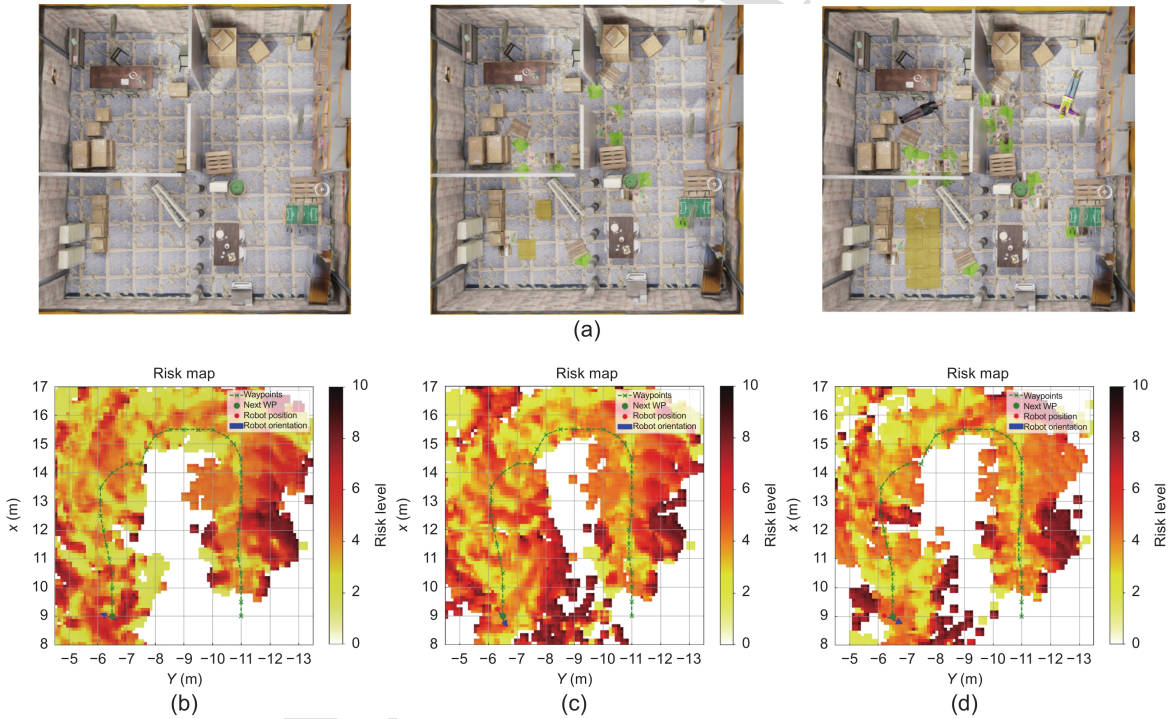


Figure 6: Generation of risk maps with different obstacle densities. (a) top view of the corresponding environment with the generated maps (b) low density (c) medium density (d) high density.

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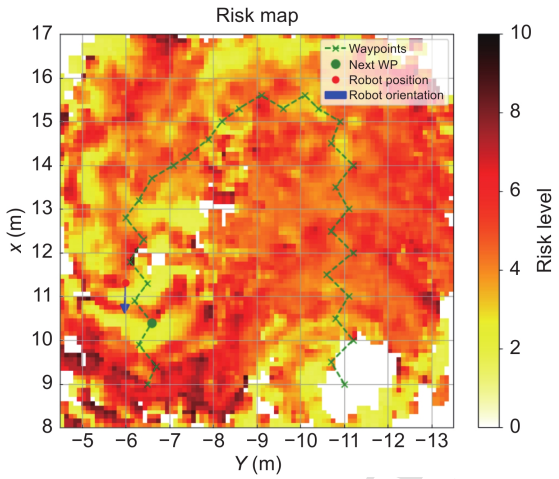


Figure 7: Wide coverage map of the environment generated using a zigzag strategy, by increasing the robot's rotation, more area is incorporated into the map.

the cost of a 44.1% increase in mission time. These results indicate a clear trade-off: zigzag movement is advantageous when maximizing environmental knowledge is critical, while normal movement remains preferable for time-sensitive operations.

4.3. Autonomous Navigation

The simulation and real test phase is available at Appendix B.

4.3.1. Simulation Stage

The risk-aware navigation framework was evaluated under varying obstacle densities and light conditions, by moving through the 4 instances of the environment in the Figure 6. Three strategies were compared: Uninformed (fixed straight-line path without risk awareness), Semi-informed (offline waypoint corrections after failures using past risk), and Informed (Real-time) (dynamic waypoint adaptation based on the risk map). The informed strategy achieved the highest completion rate (80% vs. 20% uninformed), reducing mission time by 14.7% (81.73 s vs. 95.84 s) and trajectory length by 16.0% (14.79 m vs. 17.60 m), while maintaining similar coverage ($\sim$37%) and mean speed (0.17–0.18 m/s). Risk scaling also improved stability, reducing roll/pitch oscillations and falls. Table 5 summarizes these results.

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Table 5: Performance metrics for navigation strategies in simulation (mean of 10 executions)

Metric	**Uninformed**	**Semi-informed**	**Informed (Real-time)**	**Informed (Poor-light)**
Trajectory Completion (%)	20	50	80	80
Time (s)	95.84	101.68	81.73	141.86
Distance (m)	17.60	17.26	14.79	26.40
Coverage (%)	35.54	31.70	36.77	69.43
Mean Speed (m/s)	0.18	0.17	0.17	0.19
Mean Inclination Error (°)	18.22	15.72	12.73	13.41

Figure 8(a) complements these results by visualizing the robot's partial trajectory (in green) and the corresponding risk maps generated under each strategy for a displacement between the first and second instances of the scenario. In the uninformed case, the robot follows a fixed path with minimal adaptation, resulting in fragmented risk perception and frequent failures. The semi-informed strategy shows moderate improvement, with offline waypoint corrections that slightly enhance coverage. The informed (real-time) strategy demonstrates dynamic adaptation, with the robot actively avoiding high-risk zones and generating a coherent risk map. Notably, the informed (poor-light) strategy maintains high performance despite degraded visibility, as shown in the bottom views of the figure. The left image depicts the environment under normal lighting, while the right image shows the same scene under poor-light conditions, where object detection becomes more challenging. Nevertheless, the system performs robustly thanks to the convolutional neural network's training on diverse textures and its generalization capabilities, which allow it to maintain accurate risk estimation and trajectory adaptation even under limited visual input.

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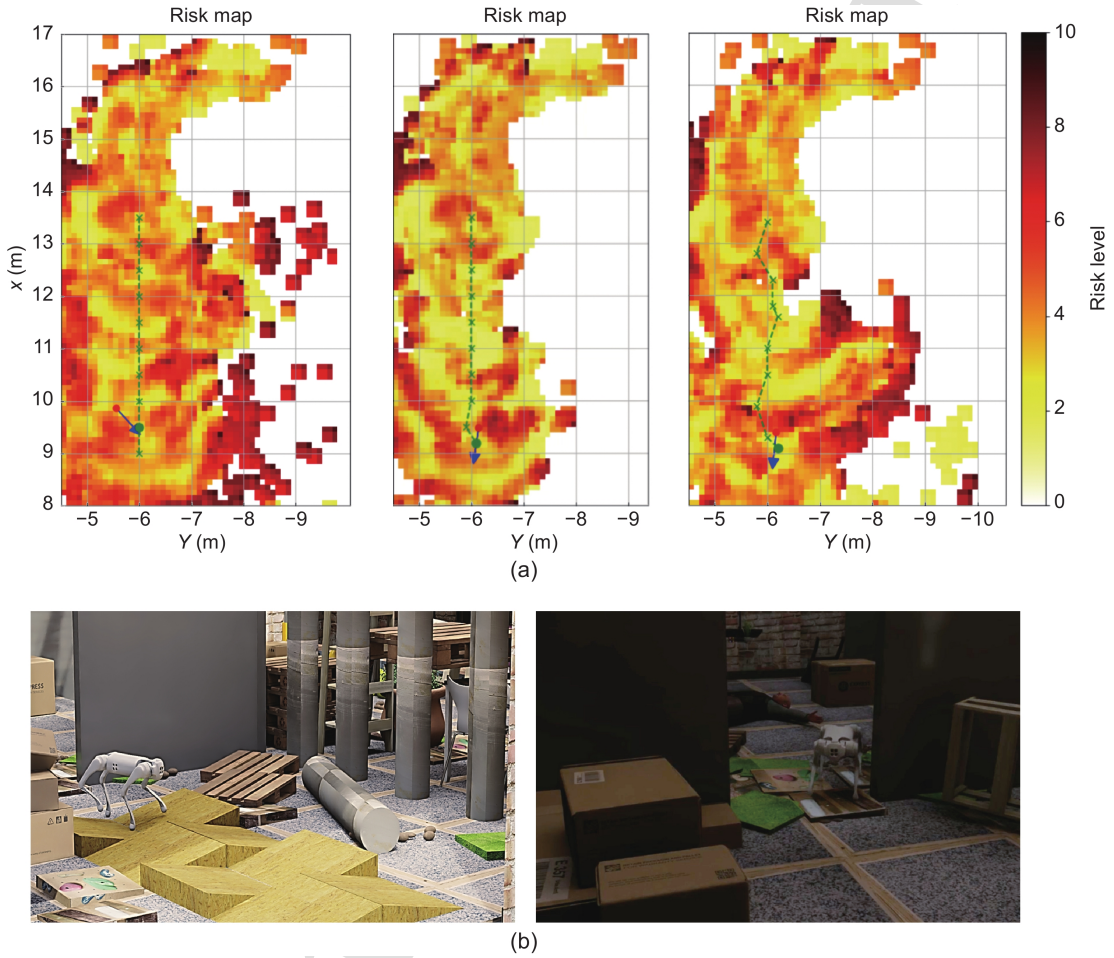


Figure 8: Simulation stage tests: The exercise consists of the robot autonomously moving from an initial zone A to a target location B within the simulated environment. (a) Risk map generated for Uninformed (left), Semi-informed (center), Informed locomotion (right). (b) Simulation movement in normal and poor light conditions.

4.3.2. Real Stage

To assess the transferability of the proposed navigation framework from simulation to physical deployment, a series of real-world tests were conducted using the Unitree quadruped robot (Aliengo and Go2) at the CAR-Robotics-Arena. These experiments aimed to qualitatively validate the system's robustness and adaptability in realistic conditions, including obstacle-rich envi-

ronments, steam interference, and variable lighting. Figure 9 illustrates two representative cases:

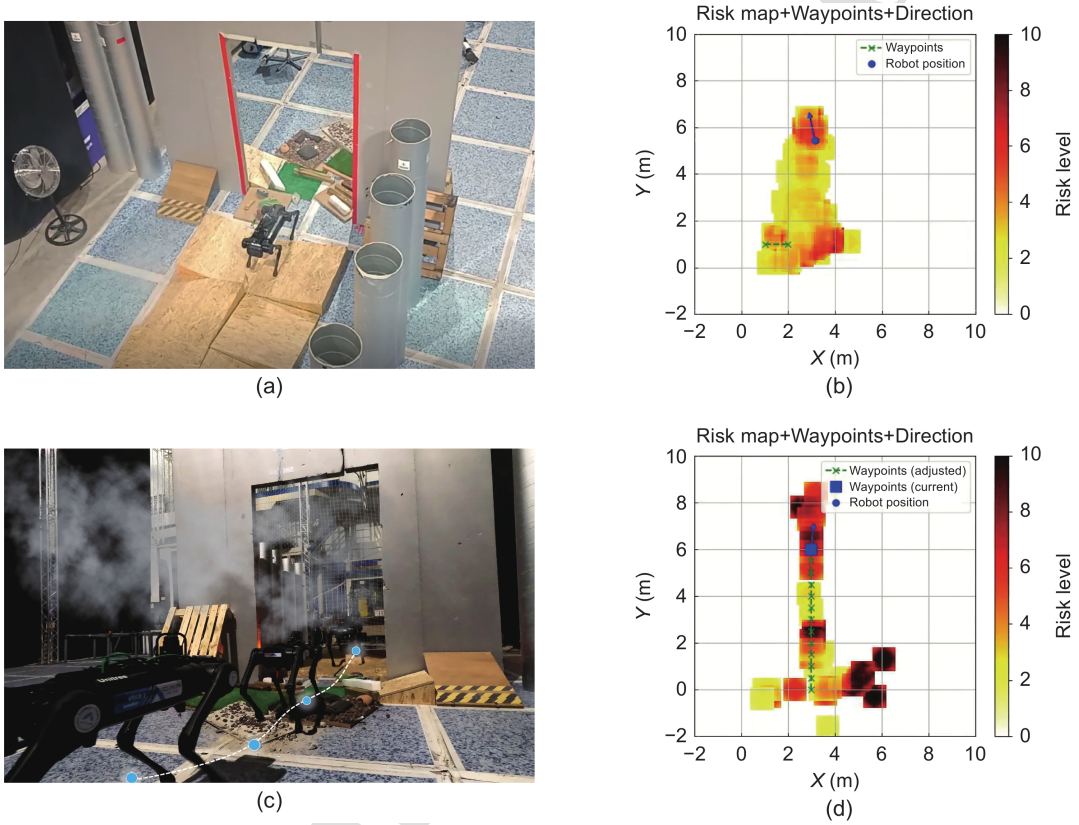


Figure 9: CAR-Robotics-Arena real tests. (a-b) Normal Light conditions. (c-d) Challenging Light conditions.

- Figure 9(a) shows the robot navigating under normal lighting conditions, where object detection is optimal.
- Figure 9(c) depicts the same environment under poor-light conditions, with reduced visibility and smoke interference.

Corresponding risk maps are shown in Figure 9 (b) and(d):

- In Figure 9(b), the risk map under normal lighting displays moderate risk levels, with well-distributed detections and waypoint adjustments.

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- In contrast, Figure 9 (d) reveals more intense color gradients, indicating higher perceived risk. This is a direct consequence of reduced visual input, which increases uncertainty in object classification and depth estimation.

Despite these challenges, the system maintained robust performance. The CNN, trained on diverse textures and lighting scenarios, demonstrated strong generalization capabilities, enabling the robot to successfully adapt its trajectory and avoiding high-risk zones even in degraded visual conditions.

Table 6: Performance Metrics in Real-World Autonomous Navigation

Metric	**Normal Light**	**Poor Light**
Trajectory Completion (%)	98	93
Time (s)	66.49	96.72
Distance (m)	5.29	6.10
Coverage (%)	28.60	69.43
Mean Speed (m/s)	0.16	0.19
Mean Inclination Error (°)	11.7	12.9

4.4. Comparative Analysis: Geometric vs. Semantic Planning

To quantify the specific contribution of the proposed bio-inspired risk framework beyond the internal strategies analyzed in Section 4.3.1, a comparative ablation study was conducted against a state-of-the-art **Geometric-Stochastic Planner**, as defined in the authors' previous work [33].

4.4.1. Baseline Definition

The comparison evaluates two distinct navigation architectures operating in the same mixed-material environment:

- **Baseline (Geometric-Only) [33]:** This framework generates optimal trajectories based on Digital Elevation Models (DEM) and stochastic sampling. It minimizes an objective function based on energy consumption, path length, and geometric slope. However, it operates with *semantic blindness*: it treats all geometrically traversable surfaces (e.g., varnished wood, loose debris, high-friction concrete) as equally safe, provided the slope is within limits.

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- **Proposed (Semantic Risk-Aware):** This system integrates the Semantic Risk Map and Anticipatory Gait Adaptation. Unlike the baseline, it assigns variable traversal costs based on visual material classification (e.g., assigning higher risk scores to `wood` or `loose debris`) and modulates swing height prior to contact.

4.4.2. Quantitative Results

Both systems were subjected to 20 experimental trials in a heterogeneous environment characterized by surfaces with varying friction coefficients and localized smoke obscuration (as detailed in Table 1). The performance metrics recorded include Success Rate, average frequency of Instability Events (defined as slippage > 5 cm or stumble recovery), average slippage distance per meter traveled, and Time to Goal (TTG).

Table 7: Quantitative Comparison: Geometric Baseline vs. Semantic Risk-Aware.

Method	**Success Rate**	**Instability Events**	**Avg. Slip** (m/m)	**Avg. TTG (s)**
Baseline (Geometric) [33]	65%	4.2 ± 1.5	0.18	**22.4**
Proposed (Semantic)	**95%**	$\mathbf{0.8 \pm 0.6}$	**0.05**	28.1

4.4.3. Analysis of Results

As evidenced in Table 7, the **Baseline** planner [33] achieves a lower Time to Goal (TTG) of 22.4 s. This reduced traversal time is a direct consequence of the geometric-stochastic optimization objective, which prioritizes the shortest Euclidean paths across surfaces that appear planar, regardless of their semantic properties (e.g., friction coefficients). However, this semantic blindness comes at a significant cost in stability: the baseline incurs a high rate of instability (4.2 events per trial) and a reduced success rate (65%), as the nominal gait controller cannot preemptively compensate for traction loss on low-friction materials such as varnished wood.

In contrast, the **Proposed Method** exhibits a temporal increase of approximately +5.7 s. It is critical to note that this additional time is not due to computational latency—the replanning loop operates at negligible cost (< 0.1 ms)—but rather stems from deliberate *conservative behavioral adjustments* triggered by the risk-aware framework:

1. **Preventive Velocity Scaling:** Upon detecting high-risk semantic classes, the high-level controller dynamically caps the maximum linear

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velocity (v_{max}) to ensure the robot operates well within the friction cone limits, preventing slippage before it occurs.

2. **Anticipatory Gait Modulation:** As defined in Eq. (6), the system proactively increases swing height (h_{swing}) in cluttered zones. While this prevents tripping, the resulting high-stepping gait inherently extends the gait cycle period compared to the nominal flat-ground locomotion used by the baseline.
3. **Risk-Aware Path Deviation:** The semantic planner rejects the shortest Euclidean path in favor of longer, non-linear trajectories that circumvent high-risk textures (e.g., loose gravel or smooth wood), naturally increasing the total path length.

Consequently, the increased TTG represents a tactical trade-off between speed and reliability. This semantic foresight reduces average slippage by approximately 72% (0.18 $\rightarrow$ 0.05 m/m) and boosts the mission success rate to 95%. These results validate that in unstructured post-disaster scenarios, optimizing for geometric traversability alone is insufficient; the semantic layer is essential to transition from "fast but fragile" locomotion to "robust and reliable" navigation.

4.5. Discussion

The results confirm that the proposed risk-aware navigation framework effectively extends quadruped capabilities beyond prior work. Unlike traditional approaches that rely on static terrain models or pre-computed maps [18, 19, 20, 21, 22], our system continuously updates risk maps in real time, enabling adaptive waypoint planning and dynamic trajectory adjustment in response to evolving hazards. The high structural correspondence between simulation and real-world features (79–87%) demonstrates that the simulation environment accurately captures critical patterns necessary for perception-driven navigation, supporting a reliable sim-to-real transfer.

Analysis of obstacle density and coverage strategies highlights a clear trade-off between exploration and efficiency. Zigzag movements increased risk map coverage by over 30% relative to straight-line navigation, illustrating the benefit of comprehensive environmental sensing for informed decision-making in cluttered terrains. These findings extend beyond previous static or semi-informed planners [27, 25], which either relied on offline semantic labeling or lacked real-time adaptation capabilities.

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Simulation experiments under varying lighting and obstacle conditions show that the informed, real-time strategy achieves superior performance across trajectory completion, mission time, and stability metrics, outperforming both uninformed and semi-informed approaches. Importantly, the CNN trained on diverse textures and lighting scenarios demonstrated strong generalization, enabling robust navigation even under degraded visual conditions. The system actively avoids high-risk zones, generating coherent risk maps that guide trajectory adaptation without modifying the underlying locomotion controller, a step forward compared to prior risk-aware or RL-based methods [20, 21].

Real-world tests validate these observations: the robot successfully navigates complex, obstacle-rich environments under both normal and poor-light conditions, maintaining high trajectory completion (98% and 93%) and effective risk-aware coverage (28.6% and 69.4%) as shown in Table 6. These results demonstrate that integrating dynamic, vision-driven risk maps with adaptive waypoint planning enables quadruped robots to operate safely and efficiently in realistic SAR scenarios, overcoming the limitations of static or partially risk-aware navigation frameworks.

5. Conclusion and Future Work

This study demonstrates the effectiveness of a bio-inspired navigation framework for quadruped robots operating in hazardous and unstructured environments typical of Search and Rescue missions. By integrating semantic risk perception with adaptive waypoint planning, the proposed approach enables real-time trajectory adjustments without modifying the underlying locomotion controller. The results obtained in high-fidelity simulations and physical trials confirm several key findings:

1. **Bio-inspired risk mapping enhances autonomous navigation:** The continuous, vision-driven risk maps allow quadruped robots to dynamically adjust their trajectories, actively avoiding high-risk zones. Compared to static or semi-informed planners, this approach improves safety, coverage, and efficiency in cluttered and dynamic terrains.
2. **Sim-to-real transfer has been robust:** High-fidelity simulation, including realistic textures, lighting variations, and physics, successfully replicates real-world conditions, as evidenced by 79–87% structural correspondence. The consistent performance across simulated and physical

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tests confirms that the framework can be reliably validated in virtual environments before deployment.

3. **Real-time adaptation increases mission success:** The informed navigation strategy, leveraging CNN-based risk estimation, achieves up to 98% trajectory completion in normal conditions and 93% under degraded visual input. Adaptive waypoint planning reduces traversal time while maintaining stability, demonstrating the benefits of real-time adaptation for challenging SAR tasks.

5.1. Future Work

A key direction for future work involves incorporating Brain-Like Neural Networks (*BLNNs*) [34] to enhance real-time decision-making in high-risk scenarios, improving adaptability and robustness in unpredictable environments. Leveraging BLNNs could allow the robot to better anticipate hazards, dynamically adjust its trajectory, and make context-aware navigation decisions in complex, rapidly changing disaster conditions. Insights gained from this research contribute to the long-term goal of deploying intelligent quadruped robots in safety-critical applications, further bridging the gap between simulation and real-world autonomy.

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Appendix A. Algorithms and environments used in IsaacSim

`https://github.com/Robcib-GIT/active_legged_learning`

Appendix B. Test execution video

Test Phase Video.

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Appendix C. Datasets and trained CNN models

`https://github.com/Robcib-GIT/EXTRA-BRAIN/tree/main/neural-network-training`

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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